

2025-11-03_R-Final_Project

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Library Installation

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidyr)
library(stringr)
library(lubridate)
```

Data Loading

Loading each monthly sales

```
jan_sales <- read_csv("data/raw/2019-sales-monthly/201901-sales.csv")
```

```
## Rows: 9723 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(jan_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>       <chr>
## 1 141234     iPhone          1              700        01/22/19 21~
## 2 141235     Lightning Charging Ca~ 1             14.95      01/28/19 14~
## 3 141236     Wired Headphones    2             11.99      01/17/19 13~
## 4 141237     27in FHD Monitor    1            149.99     01/05/19 20~
## 5 141238     Wired Headphones    1             11.99      01/25/19 11~
## 6 141239     AAA Batteries (4-pack) 1              2.99       01/29/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
feb_sales <- read_csv("data/raw/2019-sales-monthly/201902-sales.csv")
```

```
## Rows: 12036 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(feb_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>       <chr>
## 1 150502     iPhone          1              700        02/18/19 01~
## 2 150503     AA Batteries (4-pack) 1             3.84       02/13/19 07~
## 3 150504     27in 4K Gaming Monitor 1            389.99     02/18/19 09~
## 4 150505     Lightning Charging Ca~ 1             14.95      02/02/19 16~
## 5 150506     AA Batteries (4-pack) 2             3.84       02/28/19 20~
## 6 150507     Lightning Charging Ca~ 1             14.95      02/24/19 18~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
mar_sales <- read_csv("data/raw/2019-sales-monthly/201903-sales.csv")
```

```
## Rows: 15226 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(mar_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
```

```
##   <chr>           <chr>           <chr>           <chr>           <chr>
## 1 162009         iPhone             1             700           03/28/19 20~
## 2 162009         Lightning Charging Ca~ 1             14.95         03/28/19 20~
## 3 162009         Wired Headphones     2             11.99         03/28/19 20~
## 4 162010         Bose SoundSport Headp~ 1             99.99         03/17/19 05~
## 5 162011         34in Ultrawide Monitor 1             379.99        03/10/19 00~
## 6 162012         AA Batteries (4-pack) 1             3.84          03/20/19 21~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
apr_sales <- read_csv("data/raw/2019-sales-monthly/201904-sales.csv")
```

```
## Rows: 18383 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(apr_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product           'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>         <chr>           <chr>           <chr>         <chr>
## 1 176558       USB-C Charging Cable 2             11.95         04/19/19 08~
## 2 <NA>         <NA>           <NA>           <NA>         <NA>
## 3 176559       Bose SoundSport Headp~ 1             99.99         04/07/19 22~
## 4 176560       Google Phone         1             600           04/12/19 14~
## 5 176560       Wired Headphones     1             11.99         04/12/19 14~
## 6 176561       Wired Headphones     1             11.99         04/30/19 09~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
may_sales <- read_csv("data/raw/2019-sales-monthly/201905-sales.csv")
```

```
## Rows: 16635 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(may_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product           'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>         <chr>           <chr>           <chr>         <chr>
## 1 194095       Wired Headphones     1             11.99         05/16/19 17~
## 2 194096       AA Batteries (4-pack) 1             3.84          05/19/19 14~
## 3 194097       27in FHD Monitor     1             149.99        05/24/19 11~
## 4 194098       Wired Headphones     1             11.99         05/02/19 20~
```

```
## 5 194099      AAA Batteries (4-pack) 2          2.99      05/11/19 22~
## 6 194100      iPhone                  1          700.0      05/10/19 19~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
jun_sales <- read_csv("data/raw/2019-sales-monthly/201906-sales.csv")
```

```
## Rows: 13622 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(jun_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>              <chr>        <chr>
## 1 209921    USB-C Charging Cable  1              11.95        06/23/19 19~
## 2 209922    Macbook Pro Laptop    1              1700.0       06/30/19 10~
## 3 209923    ThinkPad Laptop       1              999.99       06/24/19 20~
## 4 209924    27in FHD Monitor      1              149.99       06/05/19 10~
## 5 209925    Bose SoundSport Headp~ 1              99.99        06/25/19 18~
## 6 209926    Apple AirPods Headpho~ 1              150.0        06/28/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
jul_sales <- read_csv("data/raw/2019-sales-monthly/201907-sales.csv")
```

```
## Rows: 14371 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(jul_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>              <chr>        <chr>
## 1 222910    Apple AirPods Headpho~ 1              150          07/26/19 16~
## 2 222911    Flatscreen TV          1              300          07/05/19 08~
## 3 222912    AA Batteries (4-pack) 1              3.84         07/29/19 12~
## 4 222913    AA Batteries (4-pack) 1              3.84         07/28/19 10~
## 5 222914    AAA Batteries (4-pack) 5              2.99         07/31/19 02~
## 6 222915    Bose SoundSport Headp~ 1              99.99        07/03/19 18~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
aug_sales <- read_csv("data/raw/2019-sales-monthly/201908-sales.csv")
```

```
## Rows: 12011 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(aug_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>          <chr>             <chr>        <chr>
## 1 236670    Wired Headphones    2                11.99        08/31/19 22~
## 2 236671    Bose SoundSport Headp~ 1                99.99        08/15/19 15~
## 3 236672    iPhone              1                700.0        08/06/19 14~
## 4 236673    AA Batteries (4-pack) 2                 3.84        08/29/19 20~
## 5 236674    AA Batteries (4-pack) 2                 3.84        08/15/19 19~
## 6 236675    Wired Headphones    1                11.99        08/02/19 23~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
sep_sales <- read_csv("data/raw/2019-sales-monthly/201909-sales.csv")
```

```
## Rows: 11686 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(sep_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>          <chr>             <chr>        <chr>
## 1 248151    AA Batteries (4-pack) 4                 3.84        09/17/19 14~
## 2 248152    USB-C Charging Cable  2                11.95        09/29/19 10~
## 3 248153    USB-C Charging Cable  1                11.95        09/16/19 17~
## 4 248154    27in FHD Monitor      1               149.99        09/27/19 07~
## 5 248155    USB-C Charging Cable  1                11.95        09/01/19 19~
## 6 248156    34in Ultrawide Monitor 1               379.99        09/13/19 14~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
oct_sales <- read_csv("data/raw/2019-sales-monthly/201910-sales.csv")
```

```
## Rows: 20379 Columns: 6
## -- Column specification -----
```

```
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(oct_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>              <chr>      <chr>
## 1 259358    34in Ultrawide Monitor 1      379.99      10/28/19 10~
## 2 259359    27in 4K Gaming Monitor 1      389.99      10/28/19 17~
## 3 259360    AAA Batteries (4-pack) 2        2.99      10/24/19 17~
## 4 259361    27in FHD Monitor      1      149.99      10/14/19 22~
## 5 259362    Wired Headphones      1       11.99      10/07/19 16~
## 6 259363    AAA Batteries (4-pack) 1        2.99      10/01/19 18~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
nov_sales <- read_csv("data/raw/2019-sales-monthly/201911-sales.csv")
```

```
## Rows: 17661 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(nov_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>              <chr>      <chr>
## 1 278797    Wired Headphones      1       11.99      11/21/19 09~
## 2 278798    USB-C Charging Cable  2       11.95      11/17/19 10~
## 3 278799    Apple AirPods Headpho~ 1      150.0      11/19/19 14~
## 4 278800    27in FHD Monitor      1      149.99      11/25/19 22~
## 5 278801    Bose SoundSport Headp~ 1       99.99      11/09/19 13~
## 6 278802    USB-C Charging Cable  1       11.95      11/14/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

```
dec_sales <- read_csv("data/raw/2019-sales-monthly/201912-sales.csv")
```

```
## Rows: 25117 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(dec_sales)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>       <chr>
## 1 295665    Macbook Pro Laptop  1              1700        12/30/19 00:~
## 2 295666    LG Washing Machine  1              600.0       12/29/19 07:~
## 3 295667    USB-C Charging Cable 1              11.95       12/12/19 18:~
## 4 295668    27in FHD Monitor    1              149.99      12/22/19 15:~
## 5 295669    USB-C Charging Cable 1              11.95       12/18/19 12:~
## 6 295670    AA Batteries (4-pack) 1              3.84        12/31/19 22:~
## # i 1 more variable: 'Purchase Address' <chr>
```

Combine the 12 months data using bind_rows

```
all_sales_2019_combine <- bind_rows(jan_sales, feb_sales, mar_sales,
                                     apr_sales, may_sales, jun_sales,
                                     jul_sales, aug_sales, sep_sales,
                                     oct_sales, nov_sales, dec_sales)
```

```
head(all_sales_2019_combine)
```

```
## # A tibble: 6 x 6
##   'Order ID' Product      'Quantity Ordered' 'Price Each' 'Order Date'
##   <chr>      <chr>      <chr>             <chr>       <chr>
## 1 141234    iPhone      1              700         01/22/19 21~
## 2 141235    Lightning Charging Ca~ 1              14.95       01/28/19 14~
## 3 141236    Wired Headphones 2              11.99       01/17/19 13~
## 4 141237    27in FHD Monitor    1              149.99      01/05/19 20~
## 5 141238    Wired Headphones 1              11.99       01/25/19 11~
## 6 141239    AAA Batteries (4-pack) 1              2.99        01/29/19 20~
## # i 1 more variable: 'Purchase Address' <chr>
```

EDA Process

Clean the Data

Check the Table Info

```
glimpse(all_sales_2019_combine)
```

```
## Rows: 186,850
## Columns: 6
## $ 'Order ID'      <chr> "141234", "141235", "141236", "141237", "141238", "~
## $ Product        <chr> "iPhone", "Lightning Charging Cable", "Wired Headph~
## $ 'Quantity Ordered' <chr> "1", "1", "2", "1", "1", "1", "1", "1", "1", "1", "~
## $ 'Price Each'    <chr> "700", "14.95", "11.99", "149.99", "11.99", "2.99", ~
## $ 'Order Date'    <chr> "01/22/19 21:25", "01/28/19 14:15", "01/17/19 13:33~
## $ 'Purchase Address' <chr> "944 Walnut St, Boston, MA 02215", "185 Maple St, P~
```

Counts the Number of Missing Cells (NA)

```
total_NA_cells <- sum(is.na(all_sales_2019_combine))  
cat("Total no. of missing cells: ", total_NA_cells, "\n")
```

```
## Total no. of missing cells: 3270
```

Counts the Number of Rows with at Least One NA

```
total_NA_rows <- sum(!complete.cases(all_sales_2019_combine))  
cat("Total No. of missing rows: ", total_NA_rows, "\n")
```

```
## Total No. of missing rows: 545
```

Counting Total No. of Rows Before Dropping

```
row_before <- nrow(all_sales_2019_combine)  
cat("Number of rows before dropping NA: ", row_before, "\n")
```

```
## Number of rows before dropping NA: 186850
```

Dropping NA Rows

```
all_sales_2019 <- drop_na(all_sales_2019_combine)  
# counting total no. of rows after dropping  
row_after <- nrow(all_sales_2019)  
cat("Number of rows after dropping NA: ", row_after, "\n")
```

```
## Number of rows after dropping NA: 186305
```

```
# if this match with the total no. of missing row, we can validate it  
cat("No. of dropped rows: ", row_before - row_after)
```

```
## No. of dropped rows: 545
```

Removing Duplicated Header Rows

```
all_sales_2019 <- all_sales_2019 %>%  
  # this remove the no. of repeated header rows  
  filter(`Order ID` != "Order ID") %>%  
  mutate(  
    OrderID = `Order ID`,  
    QuantityOrdered = as.integer(`Quantity Ordered`),
```



```

    PriceEach = as.numeric(`Price Each`),
    OrderDate = `Order Date`,
    PurchaseAddress = `Purchase Address`,
    ZIP = str_extract(PurchaseAddress, "\\d{5}$")
  ) %>%
  select(OrderID, Product, QuantityOrdered, PriceEach, OrderDate, PurchaseAddress, ZIP)

row_after_cleaning_repeated_header <- nrow(all_sales_2019)

cat("Total no. of repeated header rows that was dropped: ", row_after - row_after_cleaning_repeated_header)

```

```
## Total no. of repeated header rows that was dropped: 355
```

Check the Table Info & Exporting

```

glimpse(all_sales_2019)

## Rows: 185,950
## Columns: 7
## $ OrderID      <chr> "141234", "141235", "141236", "141237", "141238", "141~
## $ Product      <chr> "iPhone", "Lightning Charging Cable", "Wired Headphone~
## $ QuantityOrdered <int> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, ~
## $ PriceEach    <dbl> 700.00, 14.95, 11.99, 149.99, 11.99, 2.99, 389.99, 11.~
## $ OrderDate    <chr> "01/22/19 21:25", "01/28/19 14:15", "01/17/19 13:33", ~
## $ PurchaseAddress <chr> "944 Walnut St, Boston, MA 02215", "185 Maple St, Port~
## $ ZIP          <chr> "02215", "97035", "94016", "90001", "73301", "94016", ~

write_csv(all_sales_2019, "data/processed/all_sales_2019.csv")

```

Data Manipulation

Separating PurchaseAddress into 3 Columns (Street, City and State_ZIP)

```
all_sales_2019_splitting_and_formatted <- separate(all_sales_2019, PurchaseAddress, into = c("Street", "City", "State_ZIP"))
```

Add TotalSales & Separate City and ZIP columns

```

all_sales_2019_splitting_and_formatted <- all_sales_2019_splitting_and_formatted %>%
  mutate(
    TotalSales = QuantityOrdered * PriceEach,
    City = str_c(City, " (", str_extract(State_ZIP, "[A-Z]{2}"), ")"),
    ZIP = str_extract(State_ZIP, "\\d{5}$")
  )

```

Adding OrderDateTime, Month, and Hour Columns

```
all_sales_2019Splitted_and_formatted <- all_sales_2019Splitted_and_formatted %>%  
  mutate(  
    OrderDateTime = mdy_hm(OrderDate),  
    Month = month(OrderDateTime, label = TRUE),  
    Hour = hour(OrderDateTime)  
  )
```

Select and Check the Clean Data

```
all_sales_2019Splitted_and_formatted <- all_sales_2019Splitted_and_formatted %>%  
  select(  
    OrderID, Product, QuantityOrdered, PriceEach, TotalSales, Street, City, ZIP, Month, Hour, OrderDate  
  )  
  
glimpse(all_sales_2019Splitted_and_formatted)
```

```
## Rows: 185,950  
## Columns: 11  
## $ OrderID      <chr> "141234", "141235", "141236", "141237", "141238", "141~  
## $ Product      <chr> "iPhone", "Lightning Charging Cable", "Wired Headphone~  
## $ QuantityOrdered <int> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, ~  
## $ PriceEach    <dbl> 700.00, 14.95, 11.99, 149.99, 11.99, 2.99, 389.99, 11.~  
## $ TotalSales   <dbl> 700.00, 14.95, 23.98, 149.99, 11.99, 2.99, 389.99, 11.~  
## $ Street       <chr> "944 Walnut St", "185 Maple St", "538 Adams St", "738 ~  
## $ City         <chr> "Boston (MA)", "Portland (OR)", "San Francisco (CA)", ~  
## $ ZIP          <chr> "02215", "97035", "94016", "90001", "73301", "94016", ~  
## $ Month        <ord> Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, ~  
## $ Hour         <int> 21, 14, 13, 20, 11, 20, 12, 12, 10, 21, 11, 10, 18, 19~  
## $ OrderDateTime <dtm> 2019-01-22 21:25:00, 2019-01-28 14:15:00, 2019-01-17 ~
```

Exporting Clean Data into CSV

```
# write_csv(all_sales_2019Splitted_and_formatted, "data/processed/all_sales_2019_v2.csv")
```